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DIVP Assignment No :- 3

Q.1 Consider the hand X-ray images shown below: The image on the right was obtained by lowpass filtering the image on the left with a Gaussian lowpass filter & then highpass filtering the result with a Gaussian highpass filter. The images are of size 420 x 344 pixels & $D_0 = 25$ was used for both filter transfer functions.

(a) Explain why the center part of finger ring in figure on the right appears so bright & solid, considering that dominant characteristic of filtered image consists of edges of filtered images consists of edges of fingers & wrist bones, with darker areas in blw. In other words, would you not expect the highpass filter to render the constant area inside the ring as dark, since a highpass filter eliminates the dc term & reduces low frequencies?

(b) Do you think the result would have been different if the order of filtering process had been reversed?

→ 1. Understand the original image.

The original X-ray image contains different intensity regions
- Bones & edges have changing intensity.
- The center of ring contain relatively constant intensity region.
- The background & surrounding tissues have their own intensity values.

2. Apply Gaussian low-pass filter.

fn :- $H_{LP}(u,v) = e^{-\frac{D^2(u,v)}{2D_0^2}}$

where

$$D(u,v) = \sqrt{(u - M/2)^2 + (v - N/2)^2}$$

Since $D_0 = 25$

$$H_{LP}(u,v) = e^{\frac{-D^2(u,v)}{1250}} \quad \text{or} \quad 2\times(25)^2$$

The low-pass filter suppresses high-freq. components & preserves low-frequency components
As a result, the image becomes smoother & small details & sharp transitions are reduce.

3. Apply Gaussian high-pass filter.

$$H_{HP}(u,v) = 1 - H_{LP}(u,v)$$

$$\therefore \quad H_{HP}(u,v) = 1 - e^{-\frac{D^2(u,v)}{1250}}$$

Step 1: Examine the ring in Original image.
The ring is a small, highly contrasted object against the surrounding finger. Its brightness produces strong intensity transitions at its boundaries.

Step 2: Apply Gaussian Low-pass filter
The low-pass filter smooths the rings boundaries & spreads some of its intensity into neighboring pixels. The ring's sharp edges become blurred.

Step 3: Apply the Gaussian high-pass filter
The high-pass filter suppresses the low-freq. components of this blurred image & retains some of its more rapid intensity variations.

Step 4: Why is the center not necessarily black?
1. The center of ring looks approximately constant
2. A high-pass filter removes constant components
3. Therefore, center must become black.

Final: The ring appears bright & solid because it is a small, high-contrast object with strong boundaries. Gaussian low-pass filtering blurs & separate its intensity, while subsequent Gaussian high-pass filtering retains some of the spatial-frequency components associated with the ring & its boundaries. Their combined response can make central region appear bright & filled in.

(b)

The Originally ~~order is~~ No. The result would be same, assuming ideal linear filtering with same transfer f^n & no intermediate non-linear operations.

1. Low-pass followed by high-pass
$G(u,v) = F(u,v)\, H_{LP}(u,v)$ → apply Low pass
& $R_1(u,v) = F(u,v)\, H_{LP}(u,v)\, H_{HP}(u,v)$ → apply High pass

2. High pass followed by Low-pass
$G'(u,v) = F(u,v)\, H_{HP}(u,v)$ → apply High pass first
& $R_2(u,v) = F(u,v)\, H_{HP}(u,v) H_{LP}(u,v)$ → apply Low pass

3. Compare both results, multiplication is commutative.
$H_{LP}(u,v)\, H_{HP}(u,v) = H_{HP}(u,v)\, H_{LP}(u,v)$
$\therefore \quad R_1(u,v) = R_2(u,v)$

Take inverse fourier transform $r_1(x,y) = r_2(x,y)$

Q.2 Consider 3x3 spatial kernel that averages the four closest neighbors of a point (x,y), but excludes the point itself from the average.

a. Find the equivalent filter transfer function $H(u,v)$ in the frequency domain
b. Show that your result a lowpass filter transfer fn.

——→ The spatial filter kernel $h(x,y)$ of size 3x3 averages the four neighbors of pixel (x,y), excluding the pixel itself (centering the origin $(0,0)$ at middle of the kernel, the non-zero coefficients at positions (x,y) for $x,y \in (-1,0,1)$ are:

$$h(x,y) = \begin{cases} 1/4 & \text{for } (x,y) \in \{(0,1),(0,-1),(1,0),(-1,0)\} \\ 0, & \text{otherwise} \end{cases}$$

$\therefore$ 3x3 neighbors

$$\begin{bmatrix} 0 & 1/4 & 0 \\ 1/4 & 0 & 1/4 \\ 0 & 1/4 & 0 \end{bmatrix}$$

(a) Find frequency domain transfer function $H(u,v)$
The 2D DFT of spatial kernel $h(x,y)$ of size $M \times N$ (centered at $x,y=0$) given by

$$H(u,v) = \sum_{x=-1}^{1} \sum_{y=-1}^{1} h(x,y)\, e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

for a 3x3 spatial filter $(M=3, N=3)$:

$$H(u,v) = \frac{1}{4}\left(e^{-j\frac{2\pi u}{3}} + e^{j\frac{2\pi u}{3}} + e^{-j\frac{2\pi v}{3}} + e^{j\frac{2\pi v}{3}}\right.$$

Using Euler's identity, $\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}$

$$H(u,v) = \frac{1}{2}\left[\cos\left(\frac{2\pi u}{3}\right) + \cos\left(\frac{2\pi v}{3}\right)\right]$$

b) Proof of Lowpass Filter property.

To show that $H(u,v)$ is a lowpass filter transfer function, we evaluate its behavior at key frequency points

1. At the DC component (origin / low frequencies, $u=0, v=0$):

$$H(0,0) = \frac{1}{2}[\cos(0) + \cos(0)]$$

$$= \frac{1}{2}(1+1)$$

$$= 1$$

The filter passes low (zero) frequency signals completely without attenuation.

2. At higher frequencies away from origin:

as frequencies u & v increase toward the edges of frequency spectrum (e.g. $u = \pm 1, v = \pm 1$):

$$H(1,1) = \frac{1}{2}\left[\cos\left(\frac{2\pi}{3}\right) + \cos\left(\frac{2\pi}{3}\right)\right] = \frac{1}{2}\left(-\frac{1}{2} - \frac{1}{2}\right) = -\frac{1}{2}$$

The magnitude $|H(u,v)| = \frac{1}{2}$ at high freq. is lower than the peak value of 1 at origin, attenuating high-freq. components relative to low freq. DC component

Q.3 Consider the images shown. The image on the right was obtained by a) multiplying the image on the left by $(-1)^{x+y}$; (b) computing the DFT; c) taking complex conjugate of transform; d) computing inverse DFT & e) multiplying the real part of result by $(-1)^{x+y}$. Explain (mathematically) why the image on right appears as it does.

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→ 1. Multiply original image by $(-1)^{x+y}$

Let original image be $f(x,y)$

$$\therefore g(x,y) = (-1)^{x+y} f(x,y)$$

2. Compute The 2D DFT

$$G(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} g(x,y) e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

where x,y = Spatial coordinates

u,v = Frequency coordinates

M,N = image dimensions

$j = \sqrt{-1}$

$$\Rightarrow G(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} (-1)^{x+y} f(x,y) e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

3. Take complex conjugate of $G(u,v)$

We know, $G(u,v) = a + jb$

Then $G^*(u,v) = a - jb$ → complex conjugate

4. Compute the inverse DFT

$$g_1(x,y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} G^*(u,v) e^{j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

Using Conjugate property: $g_1(x,y) = g^*(-x,-y)$

$\Rightarrow \quad g_1(x,y) = g(-x,-y)$

Here, the image is reversed along both horizontal & vertical coordinates.

5. Multiply real part by $(-1)^{x+y}$

final image is $h(x,y) = (-1)^{x+y} g_1(x,y)$

$\Rightarrow h(x,y) = (-1)^{x+y} g(-x,-y)$

We know,

$$g(x,y) = (-1)^{x+y} f(x,y)$$

$$\therefore \quad g(-x,-y) = (-1)^{-x-y} f(-x,-y)$$

Put into final eqn: $h(x,y) = (-1)^{x+y} (-1)^{-x-y} f(-x,-y)$

$\therefore h(x,y) = (-1)^{x+y-x-y} f(-x,-y)$

So $h(x,y) = (-1)^0 f(-x,-y)$ & $(-1)^0 = 1$

$$\therefore \quad \boxed{h(x,y) = f(-x,-y)}$$